COMPANION TECHNICAL SUPPLEMENT

Technical Supplement A

Singular-Covariance Generalized Least Squares in H0DN: Support and Off-Support Projected Loss

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Draft date: 10 August 2026

Scope: synthesis of four frozen audit packages. No new H_0 estimate, causal explanation, DOI, repository record, or endorsement is asserted.

A1. Purpose and scope

This supplement documents the singular-covariance mechanism underlying the H0DN case study. It is a technical explanation of an audited numerical result, not a replacement Hubble-constant analysis. All empirical values refer to the frozen H0DN source state and the project's verified master packages. The supplement establishes four bounded points:

1. the public baseline is reproducible under its fixed representation and Moore–Penrose policy;
2. residuals strictly on covariance support retain the same target-dependent singular-Gaussian quadratic under invertible row-coordinate transformations;
3. the rounded public data vector has a nonzero component outside covariance support, and an exact invertible non-orthogonal row transformation changes the target obtained from the projected Moore–Penrose loss applied to that off-support vector; and
4. the observed change is a property of this off-support projected-loss extension, not a disagreement among the tested numerical solvers or a general failure of singular-Gaussian invariance.

It does not establish that the public H0DN value is wrong, identify the physical origin of the covariance null space, recommend a corrected covariance, or define a preferred H_0.

A2. Frozen linear model

Write the public network equations in the form

$$y=A\theta+\varepsilon,\ \mathrm{Cov}(\varepsilon)=C,$$

where $y\in\mathbb{R}^{255}$, $A\in\mathbb{R}^{255\times 64}$, $\theta\in\mathbb{R}^{64}$, and $C\in\mathbb{R}^{255\times 255}$. The audited covariance is symmetric positive semidefinite under the package's material-negative-eigenvalue rule. Its numerical rank is 183 and its nullity is 72. The design rank is 64, and the Moore–Penrose weighted normal matrix is full rank.

Under the frozen prescription,

$$W=C^{+},\ N=A^{T}WA,\ b=A^{T}Wy,\ \hat{\theta}=N^{-1}b.$$

The target H_0 is a component or deterministic transform of $\hat{\theta}$ in the public code. The audit reproduced

$$H_0=73.49875364360662\ \mathrm{km\ s^{-1}\ Mpc^{-1}},$$

$$\sigma(H_0)=0.8088000253378453\ \mathrm{km\ s^{-1}\ Mpc^{-1}},$$

$$\chi^2=117.55968168211207,\ df_{(adjusted)}=119.$$

The associated supernova absolute-magnitude quantity in the branch was

$$M_{(B)}=-19.251823407377074\pm0.022446576710565474\ \mathrm{mag}.$$

Varying the absolute pseudoinverse cutoff from 10^{-14} through 10^{-7} produced zero recorded spread in the baseline H_0. This is a fixed-representation stability result.

A3. Full-rank and on-support invariance benchmarks

Let D be any invertible row transformation and define

$$A'=DA,\ y'=Dy,\ C'=DCD^T.$$

If C is nonsingular, then

$$(C')^{-1}=D^{-T}C^{-1}D^{-1}.$$

Therefore the transformed normal matrix and right-hand side equal their original values:

$$(A')^T(C')^{-1}A'=A^TC^{-1}A,\ (A')^T(C')^{-1}y'=A^TC^{-1}y.$$

The generalized least-squares estimate and covariance are invariant. In this full-rank setting, changing row units, standardizing residuals, or applying any exact invertible coordinate map does not change the inference when all objects are transformed consistently.

The target-dependent quadratic has the same invariance when the covariance is singular but the residual is strictly on support [A2]. Let a symmetric positive-semidefinite covariance of rank r factor as

$$C=BB^T,$$

where B has full column rank. If the residual belongs to R(C)=R(B), it has a unique coordinate vector u such that

$$y-A\theta=Bu.$$

The Moore–Penrose quadratic then satisfies

$$(y-A\theta)^TC^+(y-A\theta)=u^Tu.$$

After any invertible row transformation D, the covariance factors as $(DB)(DB)^T$ and the transformed residual is $(DB)u$. Its quadratic is again u^Tu. Thus the parameter-dependent quadratic of an on-support singular Gaussian is invariant under invertible row-coordinate transformations. The H0DN finding below does not contradict this benchmark.

A4. Off-support projected Moore–Penrose loss

For a singular matrix, the Moore–Penrose inverse C^+ is defined by four equations [A1]:

$$CC^+C=C,\ C^+CC^+=C^+,$$

$$(CC^+)^T=CC^+,\ (C^+C)^T=C^+C.$$

The last two conditions encode orthogonality in the ambient Euclidean inner product. Orthogonal transformations preserve that geometry. For an orthogonal Q,

$$(QCQ^T)^+=QC^+Q^T.$$

For an off-support residual r, the quadratic $r^T C^+ r$ ignores the component in N(C) and evaluates only the Euclidean projection onto R(C). It is therefore a projected-loss extension, not the negative log-density of a singular Gaussian at that off-support point. An arbitrary non-orthogonal D changes Euclidean angles, lengths, and the projection associated with the transformed support. In general,

$$(DCD^T)^+ \neq D^{-T} C^+ D^{-1}.$$

Consequently, the projected Moore–Penrose loss applied to an off-support vector is not generally invariant under non-orthogonal congruence, even though (A,y,C) and (DA,Dy,DCD^T) are exact coordinate representations of the same linear equations and covariance bilinear form. This statement does not extend to residuals strictly on support, for which A3 establishes invariance.

This distinction is conceptual rather than merely numerical:

- **algebraic relation (F2):** the two triples are connected by a known invertible map;
- **target-inference invariance (F3):** the off-support projected-loss target is not preserved by the tested non-orthogonal map; the on-support singular-Gaussian target-dependent quadratic remains invariant.

A5. Audited diagonal row-standardization test

The branch used a positive diagonal transformation D that standardizes rows and transforms the design, data, and covariance together. The result was

$$\Delta\, H_0 = H_0\text{'}\text{-}H_0 = \text{-}0.052445422611000936 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The transformed projected-loss estimate is approximately

$$H_0\text{'} = 73.446308220995 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

with small solver-dependent final digits. The shift is about 0.065 of the published standard uncertainty. Its role in this supplement is not to claim practical or cosmological significance; it is a direct non-invariance result for the tested off-support projected-loss prescription.

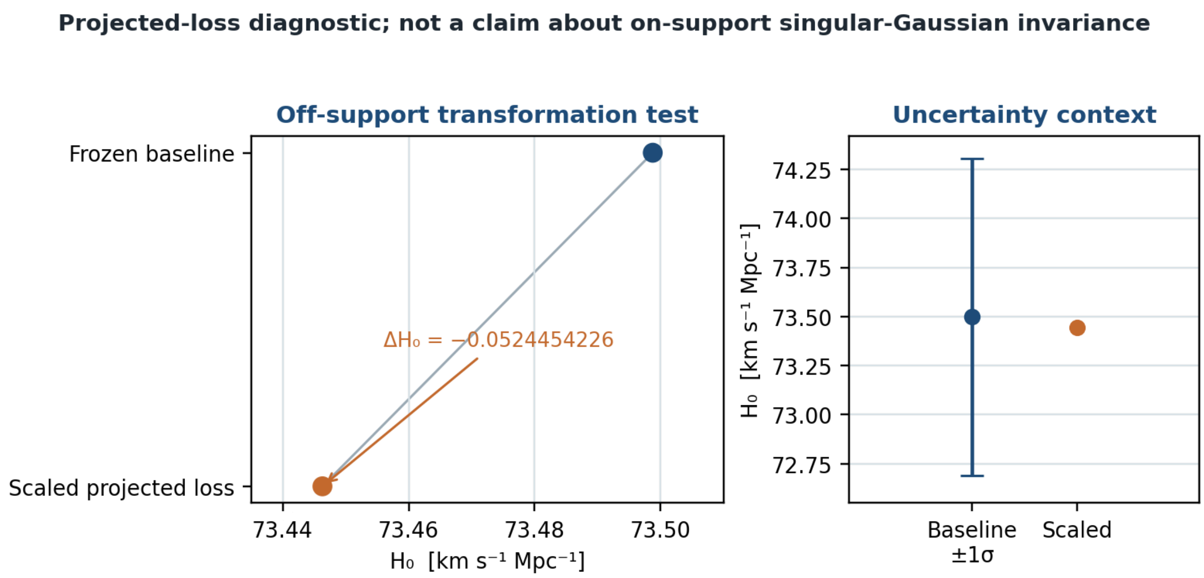

Figure A1. Frozen H0DN baseline and the consistently row-standardized off-support projected Moore–Penrose result. Error bars show the baseline published-scale standard uncertainty only to contextualize the shift; the figure is a projected-loss transformation diagnostic, not a comparison of independent H0 measurements or a statement about on-support singular-Gaussian invariance.

A5.1 Four numerical paths

Table A1. Project-internal solver checks of the non-orthogonal shift.

Path	Baseline H_0	Transformed H_0	Δ H_0
SciPy Moore–Penrose pseudoinverse	73.49875364360722	73.44630822099480	−0.05244542261242
Explicit SVD (`gesvd`)	73.49875364360699	73.44630822099464	−0.05244542261235
Symmetric eigendecomposition	73.49875364360624	73.44630822099427	−0.05244542261197
Retained-support whitening + `gelsy`	73.49875364360646	73.44630822099603	−0.05244542261043

The spread among the four Δ H_0 values is

$$1.9895196601282805\times10^{-12}\ \mathrm{km\ s^{-1}\ Mpc^{-1}}.$$

An 80-digit decimal elimination of the two 64 by 64 normal systems gave

$$\Delta\ H_0=-0.052445422611717223330578760 35\ldots$$

The agreement excludes ordinary double-precision normal-solve noise as an explanation for the effect; the SVD route follows the standard least-squares decomposition class [A3].

A5.2 Orthogonal controls

Four dense orthogonal transforms, generated under the registered seed, were applied to the actual frozen matrices. The maximum absolute change in H_0 was

$$1.5084822280186927\times10^{-10}\ \mathrm{km\ s^{-1}\ Mpc^{-1}}.$$

This is consistent with the Moore–Penrose orthogonal covariance identity. The contrast between orthogonal and non-orthogonal transformations is therefore present in the actual network matrices, not only in a toy example.

A5.3 Congruence defect

The final validation directly compared the transformed pseudoinverse with the ordinary-inverse-style congruence expression. The registered relative Frobenius defect was

$$(\|(DCD^T)^+ - D^{-T}C^+D^{-1}\|_{(F)})/(\|(DCD^T)^+\|_{(F)}) = 3.8046837415446405\times10^{-4}.$$

The nonzero defect supplies the matrix-level mechanism connecting the transformation to the changed normal equations.

A6. Exact off-support rational fixture

The mechanism can be shown without floating-point arithmetic. Let

$$C=[1\ 1;\ 1\ 1],\ A=[1;\ 1],\ y=[1;\ 0].$$

The covariance has rank one, with support spanned by $(1,1)^T$, and

$$C^+=(1)/(4)[1\ 1;\ 1\ 1].$$

For every θ, the residual is $(1-\theta,-\theta)^T$. Membership in the covariance support would require $1-\theta=-\theta$, which has no solution. The fixture is therefore explicitly off support. Its projected Moore–Penrose estimate is

$$\hat{\theta}=(A^TC^+y)/(A^TC^+A)=(1)/(2).$$

Now choose the invertible non-orthogonal scaling

$$D=[1\ 0;\ 0\ 2].$$

Then

$$C'=DCD^T=[1\ 2;\ 2\ 4],\ A'=[1;\ 2],\ y'=[1;\ 0].$$

Because $C'=vv^T$ with $v=(1,2)^T$,

$$(C')^+=(vv^T)/((v^Tv)^2)=(1)/(25)[1\ 2;\ 2\ 4].$$

The transformed estimate is

$$\hat{\theta}'=((A')^T(C')^+y')/((A')^T(C')^+A')=(1)/(5).$$

Thus an exact coordinate change sends 1/2 to 1/5 under the off-support projected-loss prescription. An orthogonal row rotation leaves the estimate at 1/2. This fixture was reproduced exactly in the final validation package. It demonstrates the same off-support extension used in the H0DN test; it is not a counterexample to the on-support result in A3.

A7. Covariance support

Let P_0 be the orthogonal projector onto N(C), the covariance null space. A degenerate Gaussian with mean Aθ and covariance C is supported on the affine space

$$A\theta+R(C).$$

Because R(C) is the orthogonal complement of N(C) for symmetric C, a literal observation y must satisfy

$$P_0(y-A\theta)=0.$$

The frozen H0DN matrices give

$$\|P_0A\|\approx1.0381406186118376\times10^{-13}$$

and

$$\|P_0 y\|_2 = 0.18874908268972845 \text{ mag.}$$

The design is numerically on the retained covariance support, while the rounded data vector contains a material null-space component. Under a literal degenerate-Gaussian model, no choice of θ can materially remove that component because P_0A is essentially zero. The H0DN transformation result therefore belongs to the off-support setting analyzed in A4 and A6.

The pseudoinverse quadratic form

$$(y-A\theta)^T C^+ (y-A\theta)$$

does not penalize the null-space component. It therefore acts as a projected loss on R(C), not as the negative log-density of a degenerate Gaussian evaluated at an off-support point. One may adopt a projected-loss convention, a rounding-tolerance convention, or a full-rank generative augmentation, but these are additional modeling choices and should be stated.

A8. Structure of the 72-dimensional null space

The audited decomposition found that the nullity equals

$$(37-1)(3-1)=72,$$

the dimension of a host-by-anchor interaction subspace after main effects are removed for 37 host levels and three anchors. The projected data component had RMS 0.0179152701 mag and maximum absolute coefficient 0.0948468468 mag, with the largest registered cell associated with the NGC 3147 by NGC 4258 interaction label.

This alignment is a structural localization, not a causal explanation. It does not show that the anchors, hosts, photometry, or covariance construction contain an error. A causal account would require the covariance-generating model, unrounded source quantities, and a documented mapping from generative components to the released matrix.

A9. Exploratory full-rank existence example

The branch included an explicitly exploratory variance-component construction to show that a full-rank model can be posed. The registered REML-like scale was

$$\tau = 0.022243618882418 36 \text{ mag,}$$

and the conditional result within that illustrative model was

$$H_0 = 73.4943259711299 \pm 0.8118670560813 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The central shift from the frozen baseline was approximately −0.00443 km s−1 Mpc−1. This calculation is an existence example, not a validated replacement covariance or corrected H0. Many full-rank augmentations are possible, and the public record does not select one uniquely.

A10. Interpretation for publication practice

When a singular covariance enters a scientific estimator, reproducible reporting should distinguish at least the following:

- the exact row representation and units;
- the rule defining numerical rank and retained support;
- the generalized inverse or constrained-solve policy;

- the class of transformations under which inference is expected to be invariant;
- the treatment of observations outside the literal covariance support;
- the generative interpretation, if any, of exact null directions; and
- sensitivity of the target to plausible representation changes.

Cutoff sweeps within one representation answer only the second item. They do not test whether the off-support projection convention is sensitive to a non-orthogonal change of row coordinates. Conversely, the observed projected-loss sensitivity should not be used to deny the on-support invariance proved in A3.

A11. Bounded conclusion

The H0DN baseline is numerically reproducible under the frozen public contract. The covariance is singular, the design is nearly in its support, and the rounded data vector has a nonzero covariance-null-space component. For this off-support vector, the adopted projected Moore–Penrose loss is stable under the tested orthogonal changes but not invariant under the tested exact non-orthogonal row scaling. Multiple solvers, high-precision arithmetic, a matrix-level congruence defect, and an explicitly off-support exact rational fixture support that bounded conclusion. Residuals strictly on covariance support retain the invariant singular-Gaussian quadratic established in A3. The results do not establish general non-invariance of singular Gaussian inference, a corrected H0, a preferred representation, a pipeline error, or a physical cause.

References for Supplement A

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